Weekly Report

Dean Reiter

March 10, 2026

Review

- Understanding how GEMini, TrigScint and GEMTrack code works
  - GEMini – GEM data processing, cluster identification and selection
  - TrigScint – Trigger data processing, trigger hit selection
  - GEMTrack – Detector track projection and selection
- Looking at GEM and trigger multiplicity in cosmic ray data versus beam data

This week

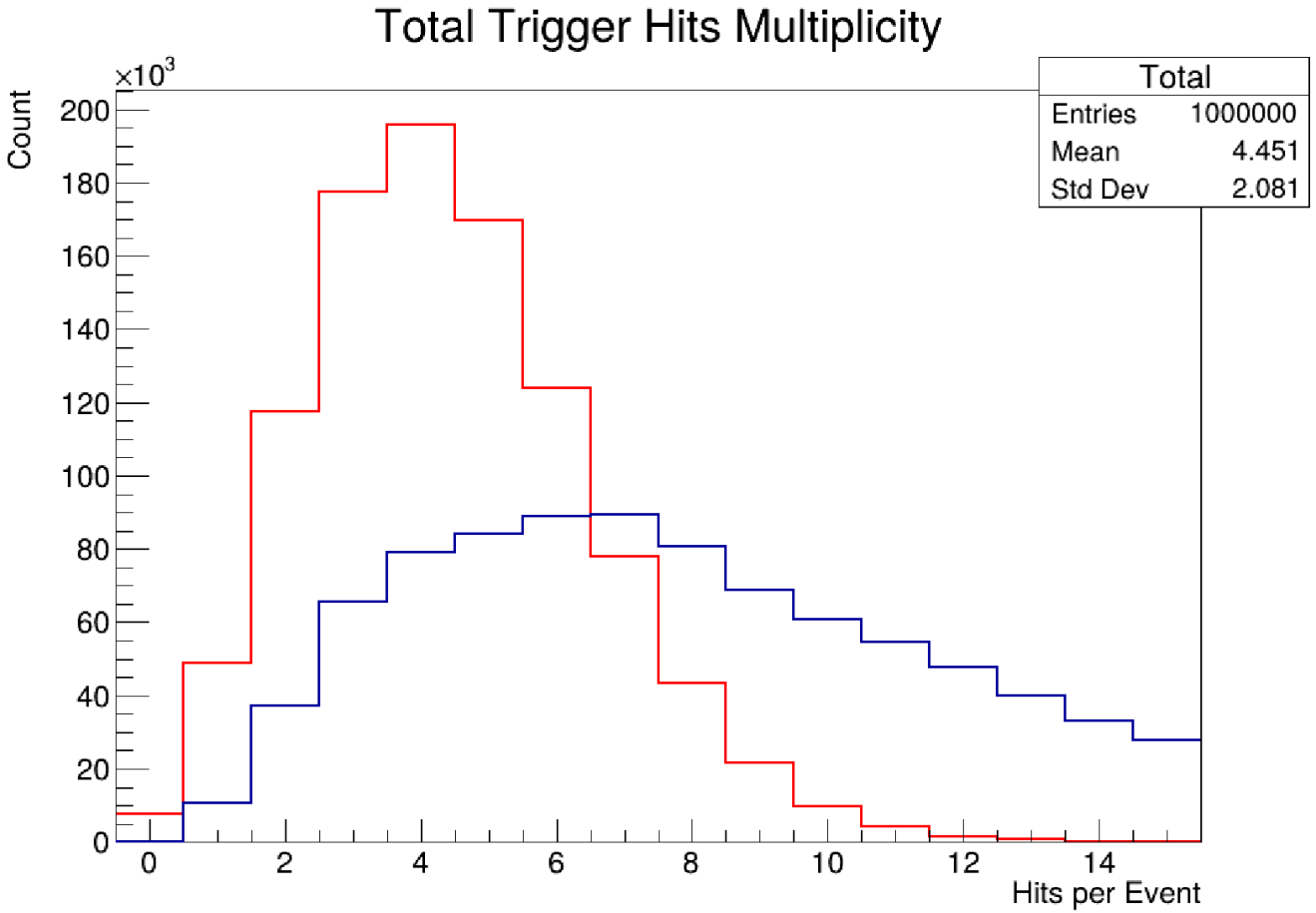
- Made some modifications to the code
- Looked at GEM and Trigger multiplicity data
  - --- Plots are a work in progress ---
  - Beam data – Run 4026
    - Jan 21, 2026 - e-e- run, 1M events, full readout, all working gems, 15 MeV, C target, 90/30 magnets
  - Cosmic data – Run 2797
    - Nov 30, 2025 - cosmic run, 1M events, full readout, all working gems

Trigger Hit Selection

- Requirements in TrigScint code:
  1. $|t_L - t_R| < 6$ ns
  2. 20 ns $< t_{L,width} < 60\ ns$ and
     20 ns $< t_{R,width} < 60$ ns

- Why?

Trigger Hit Selection



Before selection
After selection

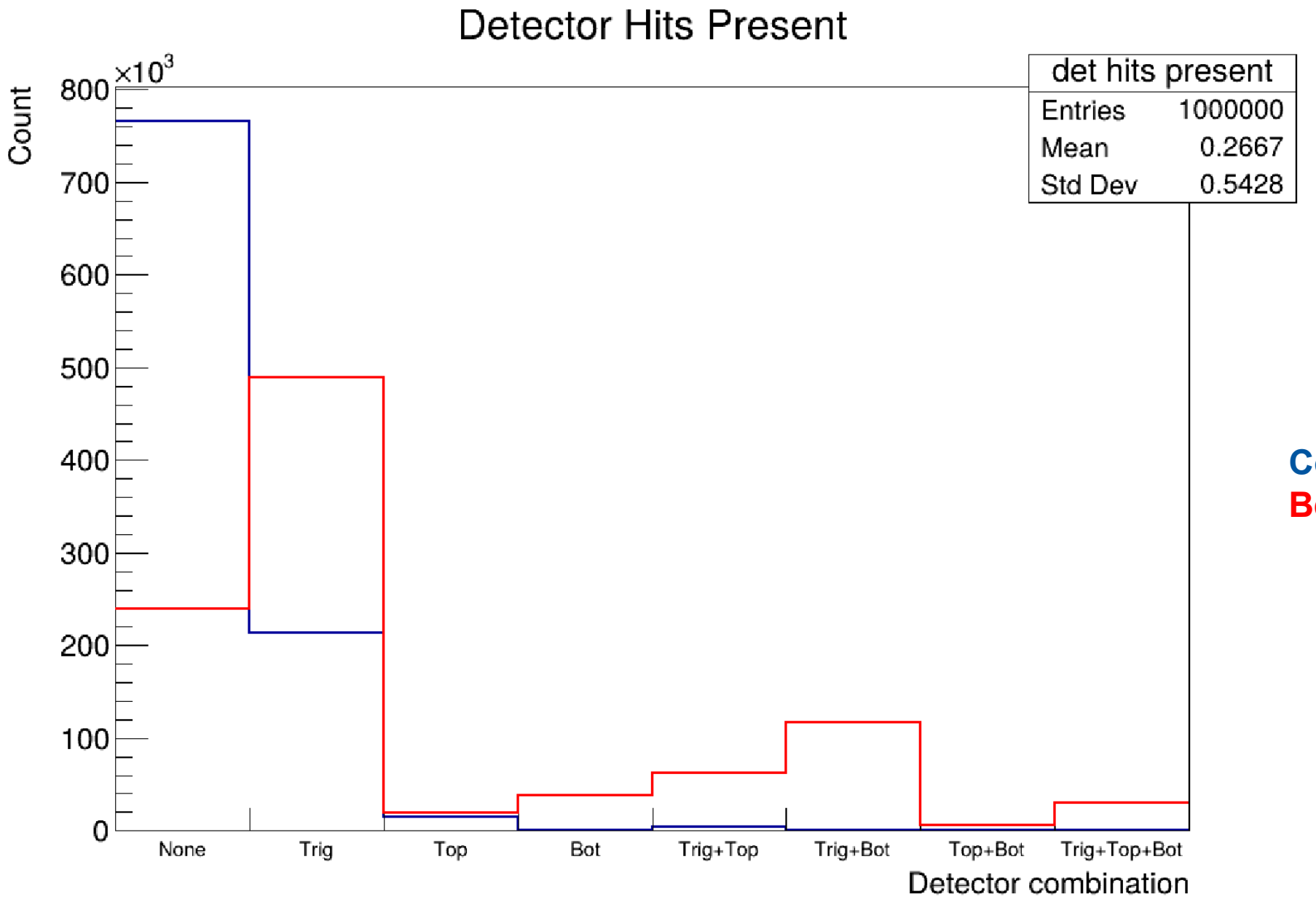
Beam data

Trigger Hit Selection

- "In-Time" versus "Out-of-time" Hits
  - Trigger hit is "in time" $|t_{hit} - t_{event}| < 5\ \mathrm{ns}$

- I removed "in time" requirement from TrigScint and GEMTrack
- I kept other requirements for $|t_L - t_R|$ and $t_{L,R,width}$

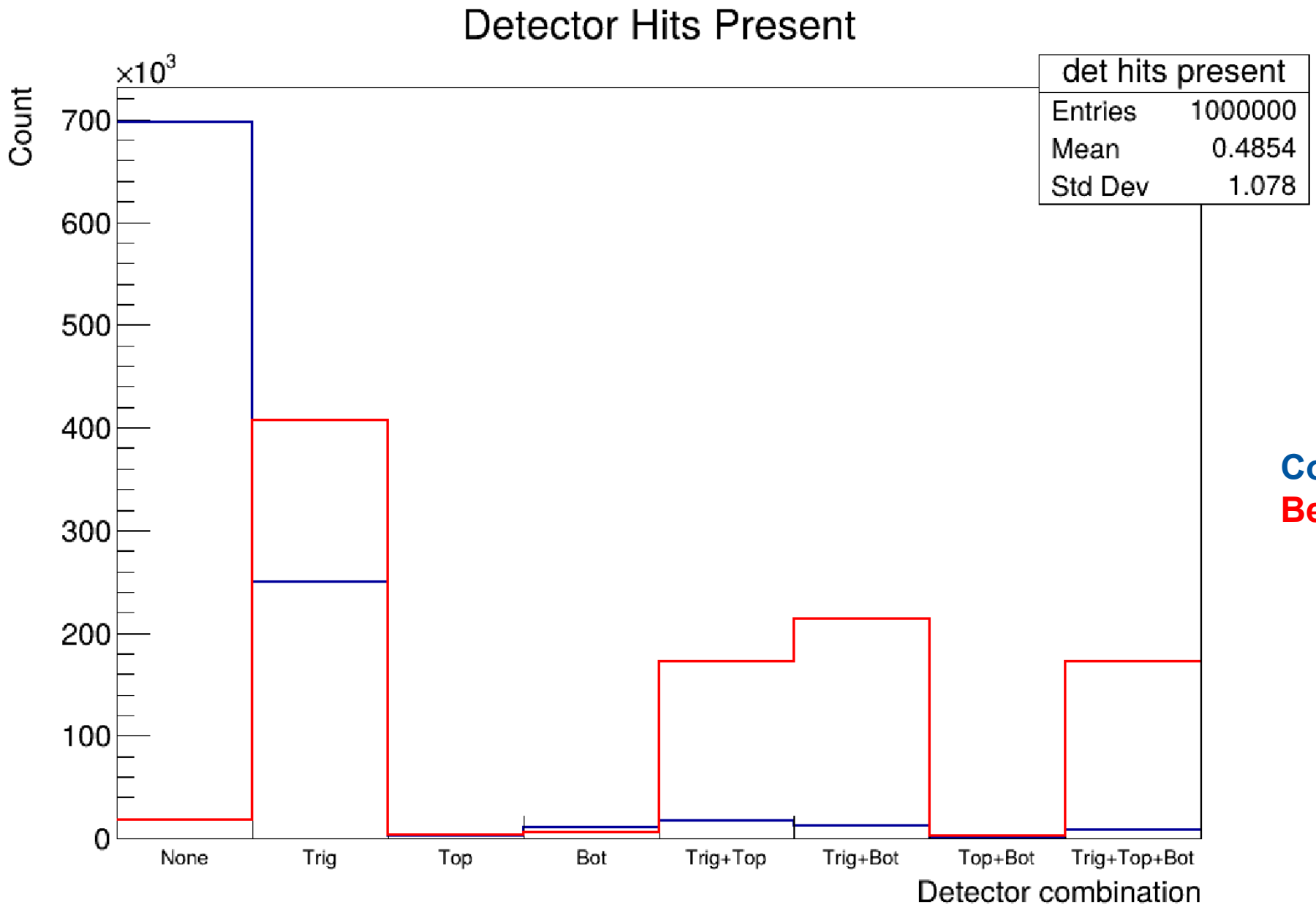
Left



Cosmic

Beam

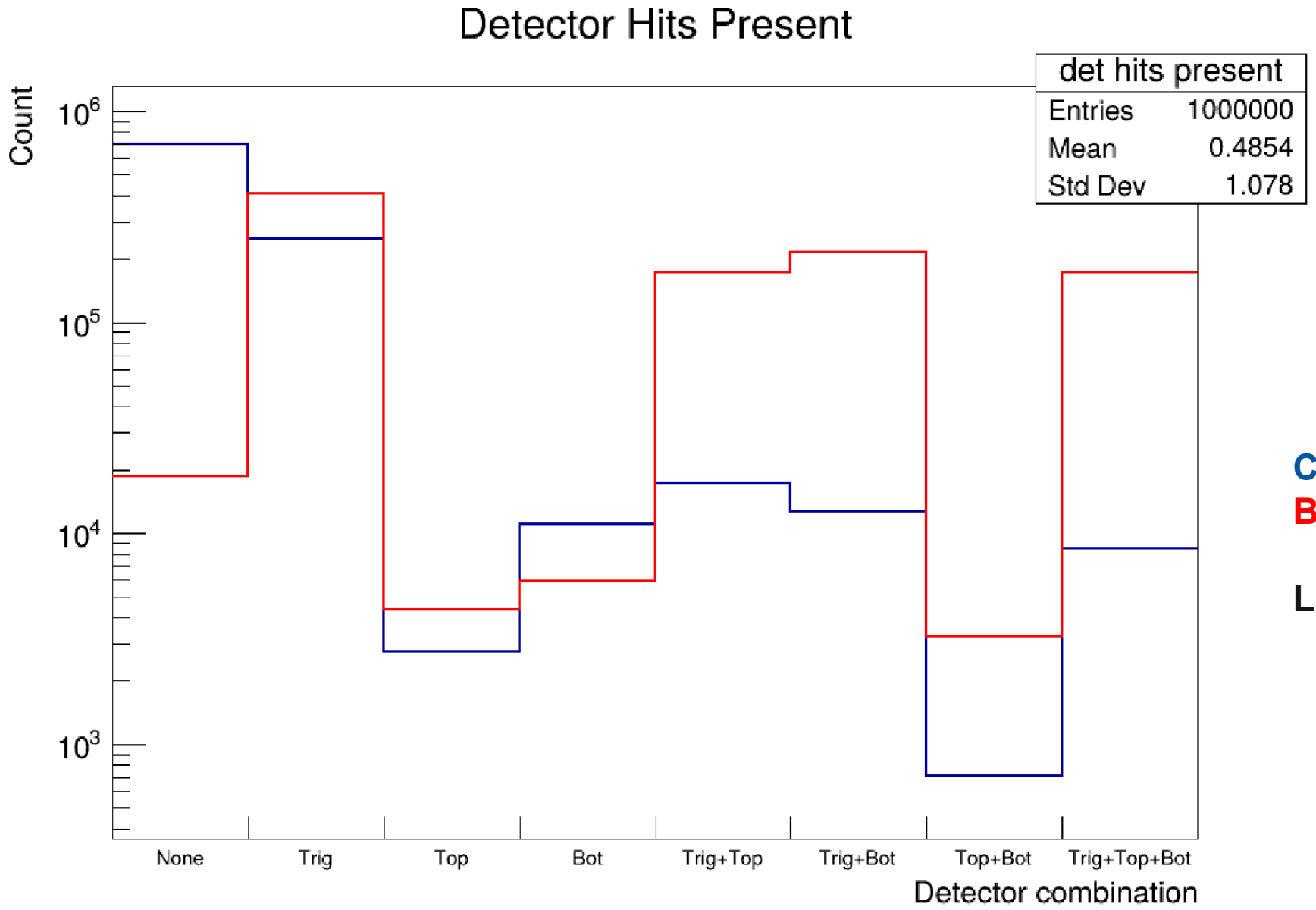
Right



Cosmic

Beam

Right

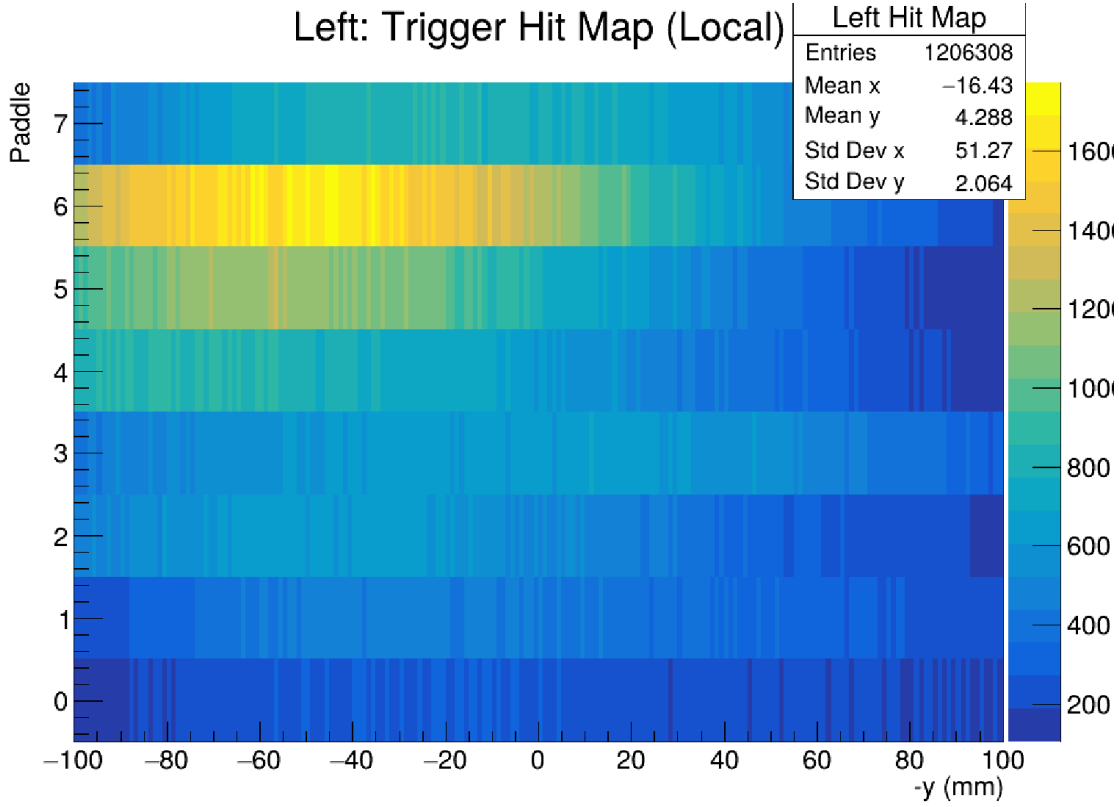


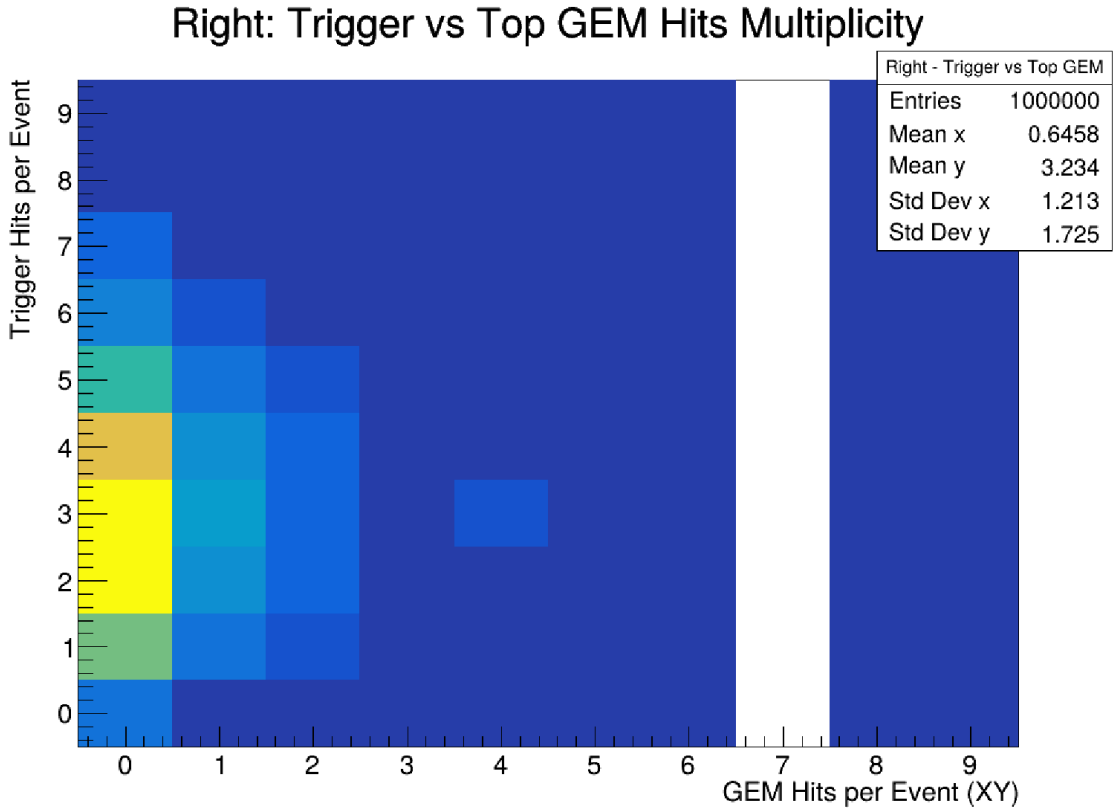
Cosmic
Beam

Log y

New Plots

- GEM vs GEM multiplicity
- GEM vs Trigger multiplicity
- Trigger Hit Maps – x (mm) vs y (mm), x (mm) vs paddle





Next Steps

- Complete study of GEM and trigger multiplicity for cosmic vs beam data
  - Look at GEM multiplicity data for various HV
- Look for issues and examine track selection in real data (cosmic, beam, simulation)
  - Think about ways to improve track selection
- Develop visual event display code
- Improve GEM cluster algorithm
  - Use timing information to correlate X and Y signals